Hunting for Exploding Stars Using Vera C. Rubin

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Honours Project A

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Rubin, LSST and Fink

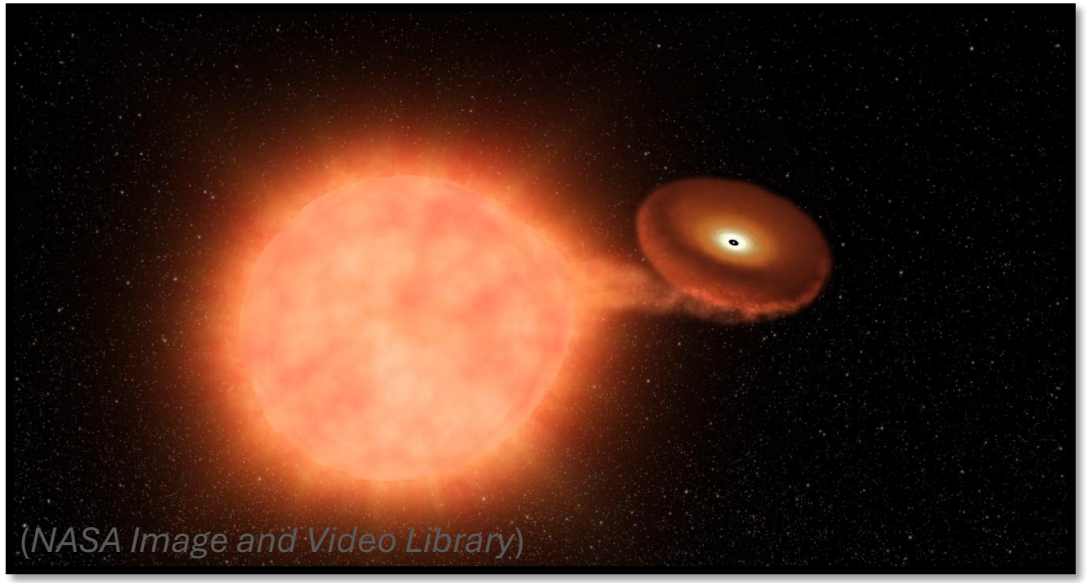
- Rubin Observatory
- 8.5 metre apertu
- Near UV, visual
- Images the sky v
  detected.
- Produce 10 mill
  10 years.
- Fink (Möller et al., 2021)
  - Filter and classify the alerts.



(Vera C. Rubin Press Release)

What are SN Ia?

- The explosion of a carbon-oxygen white dwarf (Lim et al., 2023)



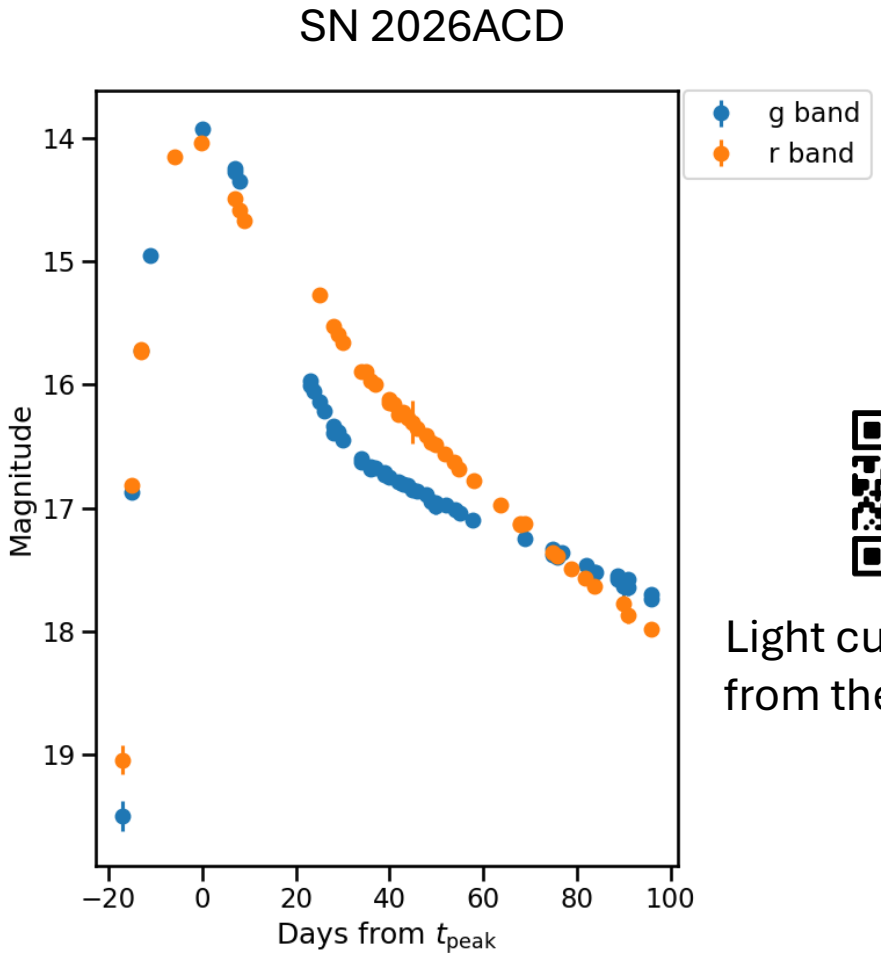
Single Degenerate Scenario



Double Degenerate Scenario

Why are SNe Ia Important?

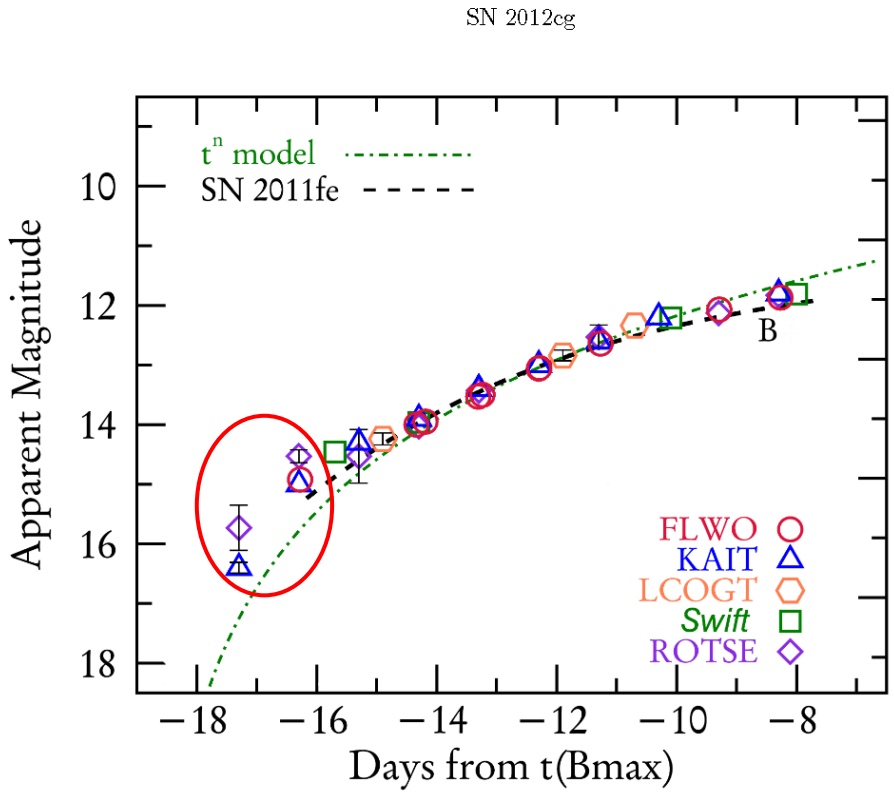
- SNe Ia are sta
  candles.
- Brighter, Bluer
  1998)
- SALT is used t
  and colour (c)



Light curve obtained from the Fink broker.

Early Excess

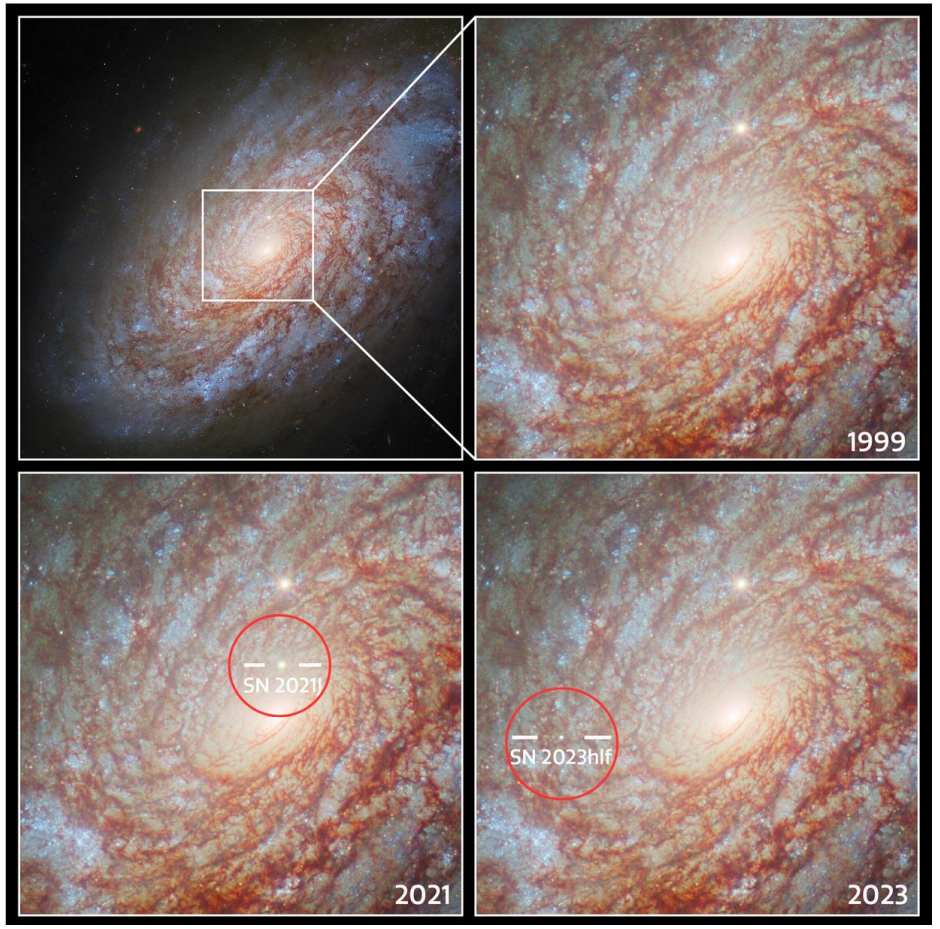
- In some cases, observation of SNe Ia have a bump in the light curve.
- Reasons For E
  - Companion 2022) and (Lim e
  - Clumping of interactions 2022).
- Colour evolut of excess leac interpretation



(Marion et al., 2016)

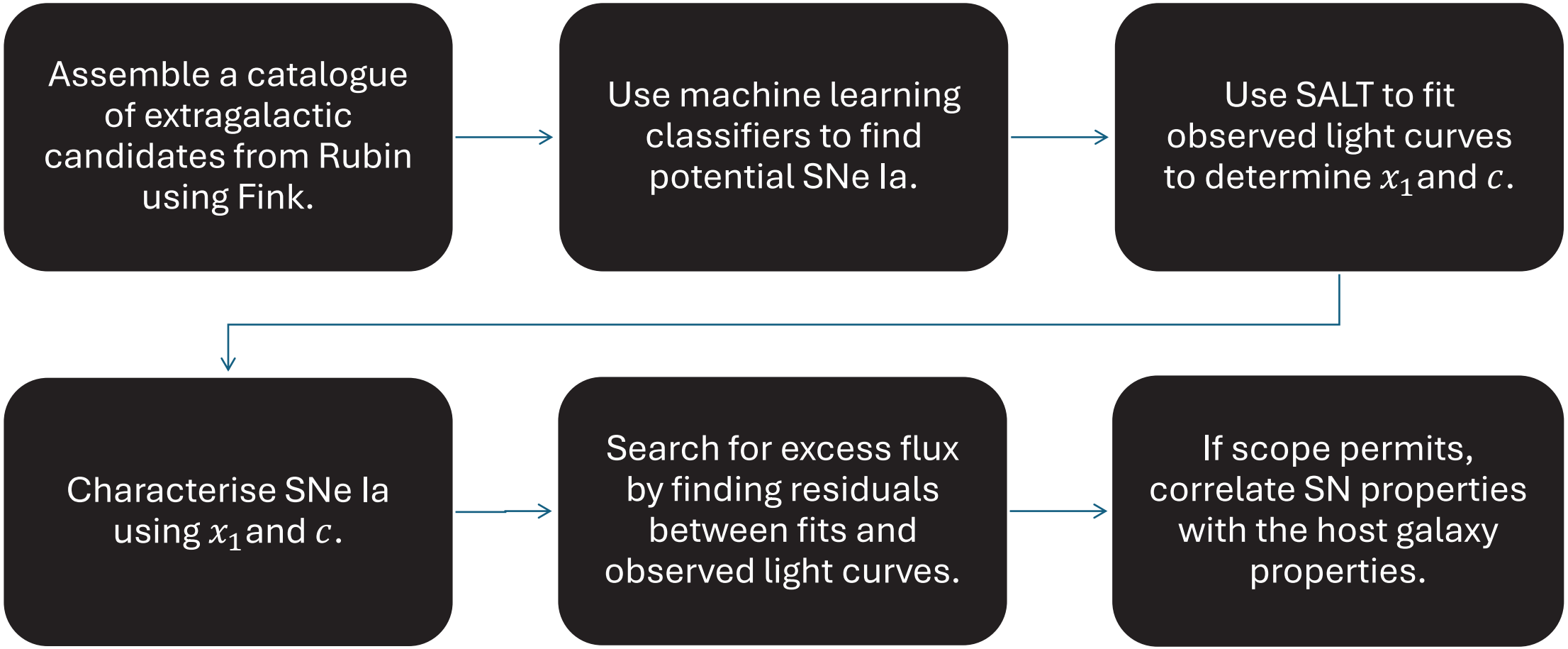
Host Galaxy Environments

- Properties
  - Stellar mass
  - Inner region and outer regions



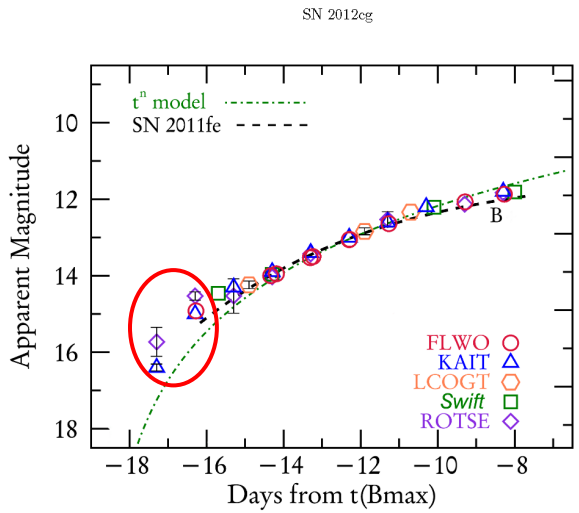
(*Supernovae in NGC 4414*, n.d.)

Proposed Method

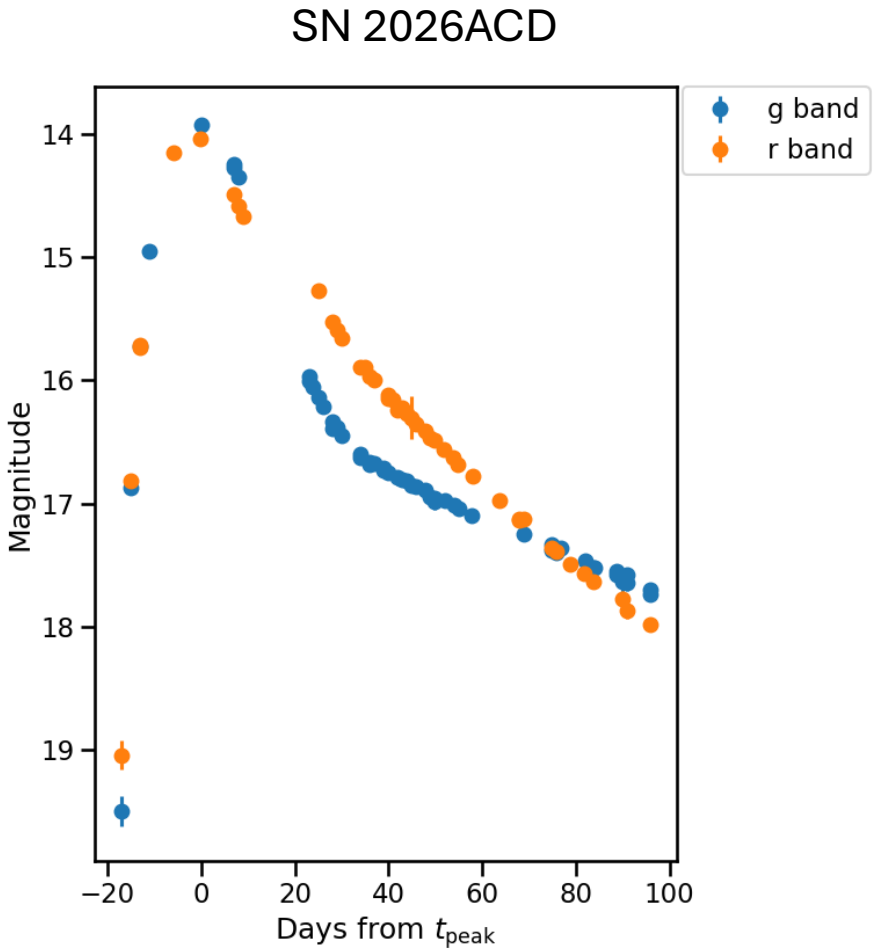


Key Takeaways

- Observe, filter and classify transients using Rubin and Fink.
- Characterise SN using SALT.
- Search for early excess SNe Ia.



(Marion et al., 2016)



Light curve obtained from the Fink broker.